# Climate Shocks and the Financial Health of American Households

Three Papers: Incidence, Persistence, and Inequality

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# Overview

Three papers on the household financial consequences of climate shocks, sharing one U.S. panel

## Three papers

- **Incidence** — do climate shocks raise the household financial burden of health?
- **Persistence** — do the effects reverse, scar, or compound?
- **Inequality** — does structural vulnerability amplify them?

## Empirical approach

- Two-way fixed effects with distributed lags; local projections for dynamics
- Natural-experiment and Callaway–Sant'Anna difference-in-differences against never-exposed counties
- Onset/persist/exit symmetry; cumulative-dose; hazard $\times$ exposure $\times$ vulnerability index
- Estimates mirrored at state and county level

U.S. panel, 2011–2023: $\sim$3,155 counties and 51 states $\times$ 13 years.

# Three Papers: Incidence, Persistence, Inequality

**Incidence (how much) → Persistence (how long) → Inequality (who bears it)**

| Paper | Question | Design | Headline |
|---|---|---|---|
| **1. Deferred Costs** (Incidence) | Do climate shocks raise the household financial burden of health? | Distributed-lag FE; never-exposed DiD; local projections | Cold (lag 1) and drought (lag 2) raise medical debt and premiums via an income channel |
| **2. Scarring & Compounding** (Persistence) | Do the costs reverse, scar, or compound? | Onset/persist/exit symmetry; cumulative-dose; Callaway–Sant'Anna | Drought debt **scars**; cold employment **compounds**; heat **saturates** |
| **3. Unequal Weather** (Inequality) | Who bears the cost? | Hazard × exposure × vulnerability (CDC SVI); EJ interactions | Harm to income, jobs, premiums is **amplified** where vulnerability is highest |

# Econometric Robustness and Methodological Additions

**Five analyses extend the core specification; each addresses a point raised in prior review**

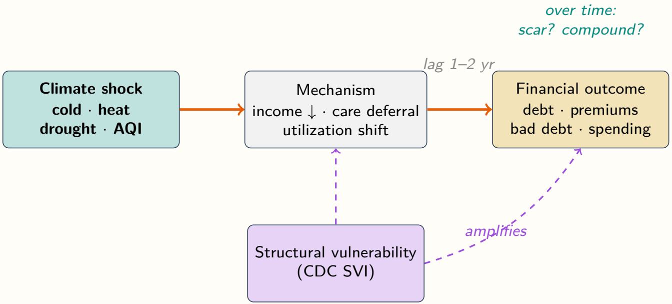
| Analysis | Result | Paper |
|---|---|---|
| Random effects (Hausman) | Rejected for all 11 outcomes ($p < 10^{-9}$); fixed effects retained | 1 |
| Post-exit / transition dynamics | Onset/persist/exit local projections; drought income partially scars | 1, 2 |
| Natural-experiment DiD | 2012 drought $2\times2$ and Callaway–Sant'Anna vs. never-exposed counties | 1, 2 |
| Propagation pathways | 18 references and descriptive figures, one set per channel | all |
| Humidity (PRISM `tdmean`) | Acquired and aggregated to state and county; headline findings stable | 2 |

# Conceptual Model and Hypotheses

What the propagation-pathway literature predicts —
stated *ex ante*, before consulting our estimates

# Conceptual Model: From Climate Shock to Health-Financial Outcome

**A lagged causal chain, moderated by vulnerability — the three papers interrogate three parts of it**




## Three questions, three papers

- **Paper 1 — the chain:** sign and lag of shock → outcome (orange arrows)
- **Paper 2 — over time:** does the effect reverse, scar, or compound? (teal)
- **Paper 3 — moderation:** does vulnerability amplify it? (purple)

One model; each paper isolates a different margin of the same chain.

# Proposed Mechanism Pathways and Supporting Evidence

## Each headline channel is anchored to peer-reviewed evidence on its direction and time-scale

| Channel | Mechanism | Key references |
|---|---|---|
| Heat → delayed care | Acute heat defers non-urgent ambulatory visits; substitution into later high-acuity ED/inpatient raises cost with a lag | Sun et al. '21; White '17; Karlsson–Ziebarth '18 |
| Cold → utilization shift | Respiratory/cardiac demand rises on impact; heating costs crowd out health spending ("heat-or-eat"), raising missed payments | Deschênes–Moretti '09; Gasparrini et al. '15; Andrews et al. '17 |
| Drought → income → debt | Farm and adjacent income falls; local consumption and ability-to-pay erode over 1–3 years — an income, not biological, channel | Hornbeck '12; Burke–Hsiang–Miguel '15; Currie–Greenstone–Meckel '17 |
| AQI → respiratory/cardiac | $PM_{2.5}$/ozone raise acute respiratory and cardiovascular events, feeding ED/inpatient utilization and cost | Deryugina et al. '19; Schlenker–Walker '16; Currie–Walker '11 |

18 references across the four channels; full set in the propagation-pathways review.

# Hypotheses: What Existing Literature Predicts

Direction and time-scale committed *ex ante* from the propagation-pathway literature — the "where should we be surprised?" lens

| Pathway | Predicted effect | Time-scale |
|---|---|---|
| Drought → income → debt | Debt ↑ (via income ↓) | lag 1–2 yr |
| Cold → utilization shift | Debt & premiums ↑ | contemp.–lag 1 |
| Heat → delayed care | Spending ↑ (deferred) | lag 1–2 yr |
| AQI → respiratory/cardiac | Utilization ↑ | within-year |

## Structural hypotheses

**H1** **Incidence:** shocks raise household financial burden with a 1–2 yr lag (income + utilization channels)

**H2** **Persistence:** drought *scars* (Hornbeck long-run); cold expected *acute* (Deschênes-Moretti) — test if cumulative exposure compounds

**H3** **Inequality:** structural vulnerability *amplifies* harm (EJ / Lancet Countdown)

## Ex-ante vs. realized

- Drought lag-2, cold lag-1 → debt: as predicted
- Long-run cold *compounding*: **stronger than predicted**
- Humidity → debt: no prior

# Paper 1 — Deferred Costs

Incidence: climate shocks raise the household financial burden of health, with a one- to two-year lag

# Paper 1: Prior Evidence from the Panel

## Two-way fixed effects with distributed lags, state and county; selected significant coefficients

| Coefficient | Estimate | |
|---|---|---|
| *State* | | |
| Employer premium: Severe Drought (lag 0/1) | +\$86 / +\$65 | ** |
| Debt share: Cold Shock (lag 1) | +0.0135 | ** |
| Total PC health exp: Cold Shock ($h$=0) | +\$205 | ** |
| Medicaid PE: Severe Drought (lag 1) | −\$525 | *** |
| *County* | | |
| Benchmark Silver: High HDD ($h$=0) | +\$22 | *** |
| Debt median (PW): High HDD ($h$=0) | +\$31 | ** |
| Hospital bad debt: High HDD ($h$=0) | +\$5.3 | *** |
| Med HH income: Z_Temp ($h$=0, heat) | −\$88 | *** |
| PCPI: PDSI (lag 2) | −\$113 | ** |

### What this establishes

- Cold and drought raise medical debt and premiums with a **1–2 year lag**
- Transmission runs through **household income**, not systemic spending
- State and county estimates agree in direction

### What it cannot settle

- Within-unit FE identifies an *association*
- Recurring shocks confound a clean before/after comparison
- Motivates a sharp natural experiment

**$p<0.05$, ***$p<0.01$; dollar outcomes in \$2023.

# The 2012 Midwest Drought: Context for a Natural Experiment

One of the most severe and extensive U.S. droughts in decades — a sharp, plausibly exogenous shock

### The event

- Severe drought across the Corn Belt and Great Plains; widespread crop losses and federal disaster declarations
- **139 counties** record their *first* extreme-drought event (PDSI $\leq -4$) in 2012 — a 4.4% onset rate
- A documented event with a defined geography and timing

### Difference-in-differences design

$$Y_{it} = \alpha_i + \gamma_t + \tau\,(\text{Treated}_i \times \text{Post}_t) + \varepsilon_{it}$$

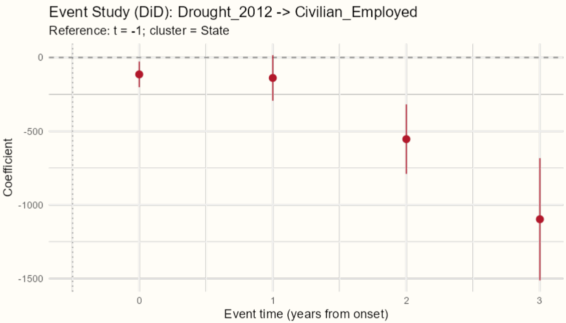
- **Treated:** first-ever extreme drought = 2012 (139 counties)
- **Control: never-exposed** counties, 2011–2023 (2,534)
- Cluster: state; event-study reference year $t = -1$

A sharp event against never-exposed controls closes the recurring-shock loophole that within-county lags cannot.

# Paper 1: What the Natural Experiment Adds

**Treated vs. never-exposed counties: income and employment fall, accumulating over the event window**

### 2012 Drought → Civilian Employed




Near-zero pre-period; post-event loss accumulates (∼-100 at $t$=1 to ∼-1,100 at $t$=3).

| 2×2 ATT | Estimate | $p$ |
|---|---|---|
| **PCPI Real** | −\$1,311 | **0.027** |
| **Civilian Employed** | −2,053 | **0.0001** |
| Med HH Income Real | −\$992 | 0.24 |
| Medical Debt Share | −0.0062 | 0.49 |

**Prior + experiment**

- Panel: drought/cold → debt and premiums (lagged, income channel)
- Experiment: **causal** income and employment loss vs. never-exposed
- Debt is null in the 2×2 (averages over post years; LP peaks at $h$=1–2)

# Paper 2 — Scarring and Compounding

Persistence: do the costs reverse, scar, or compound?
Onset/persist/exit symmetry, cumulative-dose, chronic-exposure cohorts

# Paper 2: From Levels to Dynamics

**Level effects show shocks raise costs; post-exit local projections motivate the persistence question**

## What we knew

- Cold and drought raise debt and premiums (Paper 1; levels and lags)
- Post-exit LP: drought income *partially scars* — +\$1,044 PCPI at $h$=2 ($p = 0.0002$)
- Post-exit LP: cold-shock exit brings *immediate* hospital-cost relief ($-3.1$ at $h$=0) — no within-county scarring

## Open questions

- Is the cost of *entering* a shock mirrored by the benefit of *leaving* it? (symmetry)
- Does *cumulative* exposure compound, or does the effect saturate?
- Does the answer differ by hazard?

The level/lag specification cannot separate a reversible shock from a scarring or compounding one.

# Paper 2: Two Designs for Persistence

**Onset/persist/exit symmetry, and a cumulative-shock-years dose-response**

### Transition symmetry

- Decompose each shock into **onset** ($0 \rightarrow 1$), **persist** ($1 \rightarrow 1$), **exit** ($1 \rightarrow 0$) vs. never-transitioned
- Estimate the three jointly by local projection, $h = 0 \ldots 3$
- Test $\beta_{\text{onset}} + \beta_{\text{exit}} = 0$: a nonzero sum is hysteresis (scarring or over-relief)

### Cumulative-dose response

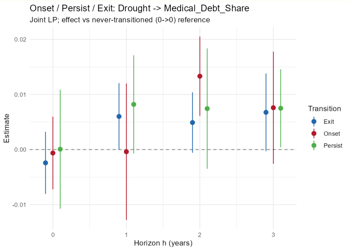
- Count **cumulative shock-years** per county (monotone, non-resetting)
- Regress outcomes on linear, quadratic, and binned dose (1–3/4–6/7–9/10+)
- Compare the marginal effect of the **10th** shock-year to the **1st**

Both estimated with county + year fixed effects, state-clustered; cross-validated against a Callaway–Sant'Anna design.

# Paper 2: Three Hazards, Three Persistence Profiles

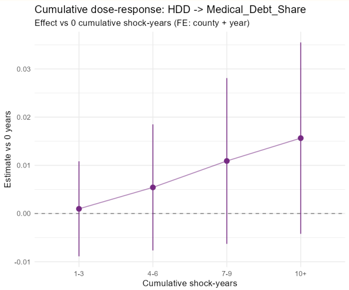
## Drought scars (onset/exit asymmetry); cold compounds and heat saturates (cumulative dose)

### Drought — scars



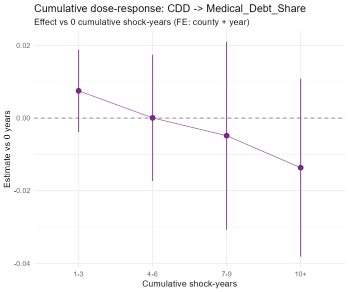

Onset and exit both raise debt at $h$=2 — not undone on exit.

### Cold — compounds




Debt share rises with each cumulative cold-year.

### Heat — saturates




Marginal debt effect flattens and reverses by year 10.

Headline: persistence is shock-specific — drought onset/exit asymmetry +0.018 ($p$=0.0015); cumulative cold raises debt to +1.5 pp at 10+ years (and compounds employment loss, −5,668 jobs);

# Paper 3 — Unequal Weather

Inequality: who bears the cost?
A hazard $\times$ exposure $\times$ vulnerability index (CDC SVI)

# Paper 3: From Average Effects to Their Distribution

**Papers 1–2 estimate population averages; who bears the cost is a separate question**

### What the averages miss

- Papers 1–2 estimate *population-average* effects of each shock
- They do not identify **which communities** absorb the cost
- A given shock may be far more costly where households are already fragile

### Hypothesis (environmental justice)

- Structural vulnerability **amplifies** climate harm
- Grounded in the environmental-justice and Lancet Countdown vulnerability-indicator literature
- Prediction: the same shock costs more at higher vulnerability

# Paper 3: A Hazard × Exposure × Vulnerability Index

Climate hazard interacted with the CDC Social Vulnerability Index; estimates mirrored at state and county

### Construction

- Combine **hazard** (shock) × **exposure** (population) × **vulnerability** (CDC SVI percentile)
- Composite scalar index, standardized
- Lancet-style **person-years** of extreme-temperature exposure

### Interaction design

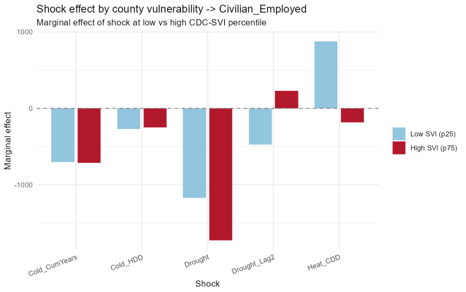
- Estimate **Shock** × **SVI** in the fixed-effects model
- Recover the marginal shock effect at **low (p25)** vs. **high (p75)** vulnerability
- Mirror every estimate at state and county level

# Paper 3: Structural Vulnerability Amplifies Climate Harm

## Shock × CDC Social Vulnerability Index; marginal effect at low (p25) vs. high (p75) vulnerability

| Shock → outcome | Low SVI | High SVI | $p_{\text{int}}$ |
|---|---|---|---|
| Heat → employment | +878 | −184 | 0.001 |
| Cold → PCPI | −\$56 | −\$472 | 0.056 |
| Drought (lag 2) → premium | −\$54 | +\$16 | 0.001 |
| Composite heat index → median income | −\$435 / **SD** | | 0.0002 |

- In vulnerable counties heat **costs** jobs; cold's income hit is ∼8× larger; drought pushes premiums *up*
- Lancet-style exposure: ∼70–105M heat person-years/yr (3–5× cold)




Heat's employment effect at low vs. high vulnerability — it flips from gain to loss.

# Credit-Bureau Medical Debt: Measurement Issues

**Income, employment, and premium results are robust; the medical-debt outcome is sensitive to how debt is recorded**

### Pattern

- Debt response concentrates in *less*-vulnerable counties
- Sign **reverses** between county and state aggregation

### Interpretation

- Credit-bureau debt requires insurance, billed encounters, a credit file
- Higher-uninsurance areas record less *measured* debt
- Consistent with a recording artifact, not absence of harm

## Cross-level evidence

- Estimates mirrored: humidity $\to$ county, SVI $\to$ state, demographics $\to$ state
- Income, employment, and premium effects **replicate** across aggregation
- Only medical debt is aggregation-sensitive

The real-economy outcomes carry the inequality result; the debt outcome is reported with the recording caveat.

# Humidity (PRISM `tdmean`) Does Not Overturn the Findings

**Mean dew point added as a control at state and county level; headline coefficients are stable**

### Procedure

- PRISM annual `tdmean` (mean dew point), 2009–2025
- Area-weighted to **state and county** polygons
- Added as a control; headlines compared with vs. without humidity on a constant sample

### Result

- State: cold (lag 1) $\rightarrow$ medical debt stable, 0.0136 $\rightarrow$ 0.0137
- County: cold $\rightarrow$ hospital bad debt and premiums essentially unchanged
- Humidity itself predicts medical debt (positive)

The drought lag-2 and cold lag-1 results are not driven by humidity confounding.

# Summary

**Findings across the three papers, and open issues**

## Findings

- **Incidence:** cold (lag 1) and drought (lag 2) raise medical debt and premiums, via a household income channel
- **Persistence:** drought debt scars; cold employment compounds with cumulative exposure; heat saturates
- **Inequality:** income, employment, and premium harm is amplified where structural vulnerability is highest

## Open issues

- Credit-bureau medical debt is measurement-sensitive
- Cumulative-CDD difference-in-differences carries a north/south regional confound — suggestive only
- Multiple testing: headline cells survive Bonferroni correction within shock

Findings are consistent across state and county aggregation throughout.

# Thank you.

Discussion and questions